# Energy & efficiency

National 5 · Engineering contexts & challenges · Worksheet

ENGINEERING WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ describe energy conversions in a system and draw an energy audit
- ☐ calculate kinetic, potential, electrical and heat energy
- ☐ calculate power, work done and efficiency
- ☐ explain ways of reducing energy losses

## I DO — Worked example

Read how a full-mark answer is built

*The lift motor uses 95 kJ of electrical energy to raise the car (Ep = 76 440 J).*

**1 Calculate the efficiency of the lift.** (2 marks)

**Full-mark answer** each ✓ is one mark

η = useful energy out ÷ energy in ✓ = 76 440 ÷ 95 000 = 0.80 or 80% ✓.

**Why marks are not awarded**

*"η = 95 000 ÷ 76 440 = 1.24"*

0 marks: values upside down. Efficiency can't be more than 1 (100%).

## WE DO — Guided practice

Use the starters and word bank to help you

*A lift raises a 650 kg car through 12 m.*

**2 Calculate the potential energy gained by the car.** (2 marks)

**Sentence starters**

One … is …, which means …

Another … is …, so …

**Word bank**

Tip: 2 marks = 2 separate, developed points.

**YOU DO** **On your own** No notes. Check your answers afterwards.

### 3 Quick check: tick one box for each.

**a)** A motor takes in 2000 J and gives 1500 J of useful work. Its efficiency is...
☐ 3500 ☐ 1.33 ☐ 0.75 ☐ 500

**b)** A 24 V, 0.42 A device runs for 7.5 hours. Which time value must be used?
☐ 27 000 ☐ 450 ☐ 7.5 ☐ 0.125

**c)** Unwanted energy in a gearbox is mainly lost as...
☐ Light ☐ Chemical energy ☐ Potential energy ☐ Heat and sound

**d)** The unit of power is...
☐ Watt ☐ Newton ☐ Joule ☐ Volt

### 4 Key terms: write the letter of the matching description next to each term.

| | | |
|---|---|---|
| 1. **Work done** | ____ | **A** $E = F \times d$ |
| 2. **Potential energy** | ____ | **B** $E = c \times m \times \Delta T$ |
| 3. **Kinetic energy** | ____ | **C** $E = V \times I \times t$ (t in seconds) |
| 4. **Electrical energy** | ____ | **D** $E = m \times g \times h$ ($g = 9.8$) |
| 5. **Heat energy** | ____ | **E** $E = \frac{1}{2} m v^2$ |
| 6. **Power** | ____ | **F** $P = E \div t$ |

*An e-bike and rider have a total mass of 110 kg and travel at 7 m/s.*

### 5 Calculate the kinetic energy. (2 marks)

*A kettle element is rated at 2.2 kW.*

### 6 Calculate the energy used in 3 minutes. (2 marks)

*1.5 kg of water is heated from 15 °C to 100 °C. Specific heat capacity of water = 4180 J/kg °C.*

**7 Calculate the heat energy needed.** (2 marks)

**Check and reflect**

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?